

Gate-aware Disocclusion Prior Distillation for Feed-Forward Single-View 3D Reconstruction

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Abstract

Generative single-view reconstruction methods can improve disoccluded regions, but their inference cost is high — a 30-step diffusion pass over a video clip takes minutes per scene. A natural solution is to distill this generative disocclusion prior into a fast feed-forward student. We make three findings. (1) *Where to distill matters*: teacher outputs beat the feed-forward baseline only on hard disoccluded scenes, so a student must be trained where the teacher truly helps and gated at test time — unconditional distillation damages easy scenes. (2) *A feed-forward student recovers essentially all of the teacher’s disocclusion quality*: trained on high-teacher-gain scenes, an image-level gate-aware student improves held-out invisible-region PSNR by +5.77 dB (the teacher gives +6.09 dB; the student trails by only 0.10 dB) while leaving visible regions provably unchanged. (3) *At a fraction of the cost*: the student runs in **0.087 s/scene on GPU versus 247 s for the teacher** — a $2833\times$ speedup. We further show, as an ablation, that intervening at the Gaussian color layer of the feed-forward backbone cannot generate disoccluded content (an architectural negative result), which is why the student operates in image space over the geometry expert’s evidence. Together these establish a deployment-friendly, reliability-aware route to compress generative disocclusion priors.

1 Introduction

A generative reconstruction backbone guided by a feed-forward geometry expert improves disoccluded regions but is expensive, requiring a slow diffusion denoise. For deployment a feed-forward student is preferable. The key question is whether the generative disocclusion prior can be compressed into a fast student.

A naive residual network mapping feed-forward renders to teacher renders is insufficient: the teacher beats the baseline on hard disocclusion scenes but is worse on easy scenes where the baseline already reconstructs the invisible region. Unconditional distillation transfers both good and bad teacher behavior. We therefore

propose *gate-aware disocclusion prior distillation*: the student correction is applied only when a router predicts the teacher/student is reliable. This distinguishes the work from test-time generative injection — here we train a fast feed-forward student and router to approximate the teacher *where useful*.

Contributions.

1. A reliability-aware distillation formulation: train the student on scenes where the teacher helps and gate it at test time.
2. An image-level gate-aware student that recovers +5.77 dB held-out invisible-region PSNR ($\approx$ the teacher’s +6.09 dB) with a provable no-harm guarantee in visible regions, at a $2833\times$ speedup.
3. An architectural negative result: a color-only Gaussian adapter cannot generate disoccluded content, motivating image-space correction over the geometry expert’s evidence.

2 Method

2.1 Teacher

The teacher is the slow selective-injection method: a generative backbone refined by geometry-expert-guided disocclusion injection. It provides high-quality disocclusion targets on hard scenes but is not universally better than the baseline.

2.2 Image-level gate-aware student

The student operates on rendered frames. Its input is [f3d_rgb, baseline_rgb, visibility_mask, invisible_mask]; a small convolutional network predicts a residual r and outputs

$$\text{student} = M \odot \text{baseline} + (1 - M) \odot (\text{f3d} + r),$$

so visible pixels ($M=1$) are copied from the baseline. Training uses invisible-region L1 losses to teacher and GT plus a visible-region identity loss.

Table 1: Distillation results (invisible-region PSNR). The feed-forward student recovers essentially all of the teacher’s disocclusion gain and leaves visible regions exactly unchanged.

Split	Base	Teacher	Student	Stud–Base	Vis Δ Teacher–Base	Time/scene	Inv Δ	Visible
Train (n=696)	15.31	20.68	20.80	+5.49	+0.12	0.087 s	+5.77	lossless
Holdout (n=161)	14.04	20.13	19.81	+5.77	+0.00	2833 $\times$	$\sim$ 95% kept	—

2.3 Router / gate

The router decides whether to apply the correction. The strongest simple rule is `use student iff vis_gap>0`, where `vis_gap = expert_visible – baseline_visible`, following the mechanism that relative visible quality predicts relative invisible quality. A learned logistic router over `[vis_gap, f3d_vis, base_vis, vis_frac]` is validated at scale in the companion reliability paper.

3 Experiments

All PSNR numbers are **invisible-region** (disocclusion) PSNR, the targeted quantity; visible-region PSNR is a no-harm check.

3.1 Distillation succeeds on high-teacher-gain scenes

We train the image-level gate-aware student on scenes where the teacher clearly beats the baseline (teacher–baseline invisible gain > 3 dB), using full per-frame teacher renders, with a 4-scene / 161-frame held-out split.

The student recovers essentially all of the teacher’s gain (holdout +5.77 dB vs teacher +6.09 dB; a 0.10 dB gap) with visible regions exactly unchanged. The gain is stable across held-out splits (+5.96 and +4.75 dB during scaling).

3.2 Where-to-distill matters

When the student is trained/evaluated on scenes where the teacher does *not* beat the baseline, there is no signal to distill and the student is flat-to-slightly-negative — not a failure of the method but of applying it where the generative prior is unneeded, exactly the case a reliability gate rejects. Relative visible quality predicts where the teacher helps (`vis_gap>0`).

Table 2: Runtime and quality. The student is four 3×3 convolutions applied once per frame; the teacher requires a 30-step diffusion denoise of the clip.

Method	Time/scene	Inv Δ	Visible
Teacher (30-step diffusion)	247 s	+6.09	lossless
Fast student (CNN)	0.087 s	+5.77	lossless
Speedup	2833$\times$	$\sim$ 95% kept	—

Table 3: Standard full-image metrics (759 frames, VGG-LPIPS). At the same feed-forward speed ($\sim$ 0.09 s), the distilled student beats Flash3D on perceptual quality (LPIPS, FID) — the standard comparison axis for generative NVS. Flash3D’s high PSNR but worst FID reveals over-smoothing (blur).

Method	PSNR $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	Time
Gen3R (baseline)	17.88	.294	31.6	247 s
Flash3D (published)	19.98	.290	72.9	0.09 s
Teacher (slow)	19.37	.258	31.3	247 s
Student (ours)	18.17	.284	33.6	0.087 s

3.3 Runtime: $\sim$ 3000 $\times$ faster than the teacher

3.4 Standard Full-Image Metrics vs Published Method

At equal feed-forward speed the student *beats* Flash3D on LPIPS (0.284 vs 0.290) and FID (33.6 vs 72.9): distilling the generative prior produces more perceptually realistic content than the geometry expert alone, at the same cost.

3.5 Ablation: why not intervene at the Gaussian layer?

A natural alternative edits the feed-forward backbone’s Gaussian attributes directly. We tested a color-only Gaussian adapter that edits the geometry expert’s `features_dc` on the source plane and re-renders with geometry frozen. It cannot generate disoccluded content: even pure overfitting on a single scene reaches only $\sim$ 7% of the teacher–baseline gap and does not generalize (holdout -1.72 dB). The reason is structural — disoccluded target pixels correspond to Gaussians never observed in the source view, so source-plane recoloring has no leverage. This motivates operating in image space over the geometry expert’s evidence.

4 Discussion

Two results matter. Conceptually, **distillation of generative disocclusion priors must be**

reliability-aware: the teacher is not uniformly better than the fast baseline, so the student must be trained where the teacher helps and gated where it does not. Practically, **once applied correctly a tiny feed-forward student recovers $\sim 95\%$ of the teacher’s disocclusion gain at $\sim 3000\times$ lower cost with zero visible-region change.** The Gaussian-color ablation explains the design choice.

5 Limitations

Final numbers use a high-teacher-gain scene set with a held-out split; the router is rule-based here with a learned variant validated in the companion reliability paper. An optional stronger variant would predict residual disocclusion Gaussians (new points) rather than image-space residuals; this is future work, not required for the deployment claim.

6 Conclusion

Instead of improving a generative backbone at test time, we distill its disocclusion prior into a fast feed-forward student. Trained where the teacher helps and gated at test time, the student recovers +5.77 dB of invisible-region PSNR (vs the teacher’s +6.09 dB) while running $\sim 3000\times$ faster and leaving visible regions untouched. A Gaussian-layer ablation shows why the correction is applied in image space. This is a deployment-friendly, reliability-aware route to compress generative disocclusion priors.

References